

EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelors in Computer Science; Statistics; Mathematics (Triple Major) GPA: 3.9

Jun 2020 – May 2024

Organizations: CS Honor Society Officer, CSUA (CS Club Executive Officer), CSM (CS 61A/B/70/88 Tutor)

Relevant Coursework: Algos, Compilers, Data Mining*, Full Stack Web[†], LinAlg, Multivar Calc., Parallel Systems[†],

Probability[†], Signal Processing, Robotics[†], Computer {Architecture, Graphics[†], Security, Vision*}, UI/UX, Neural Nets[†],

Deep RL[†], NLP[†], Causal Inference*, Forecasting[†], Real Analysis*, Theoretical Stats*[†] [†]= Graduate * = In Progress

EXPERIENCE

• Schonfeld (Quantitative Hedge Fund)

New York, NY

Applied Machine Learning Engineer (Alpha **Quantitative Researcher**) Intern

Apr 2023 - Aug 2023

- Exploratory Data Analysis using **Databricks** PySpark >100 GB of alternative data from AWS S3 and EC2, with storage optimized via Parquet files, for a statistically backtested signal. Under **Systematic Trading** team (QIS).
- Wrote an Online Algorithm to improve latency by >50% via Statistical Estimation for a **HFT Stat Arb** strategy. Measured effects of out-of-sample distributional shift via **divergence** measures. Wrote scheduler to ingest live data.

• Arrowstreet Capital LP

Boston, MA

Software Developer (Quant Dev) Intern @ Arrowstreet's **Research Core Team**

Jun 2022 - Aug 2022

- Fine-tuned a local **FinBERT** model from **Hugging Face** on over 100,000 vendor filings of various languages & correlated sentiments with word embeddings using **Cosine Similarity**, Jaccard Index, and Minkowski Distances.
- Wrote distributed code with **Numba** and **Dask** to discover hidden signals for equity alpha research. Made robust improvements to the firm-wide single systematic process with equities toolings for Bayesian Regressions.

• Electrical Engineering & Computer Sciences @ University of California, Berkeley

Berkeley, CA

Teaching: Head TA, uGSI, Tutor, Reader; Researcher, Research Assistant

Jan 2021 - Present

- Head Teaching Assistant - CS C100: Data Science and Machine Learning:** Managed admin logistics for ~300 students related to the EdStem communication platform, assignment extensions (automated using Gradescope API), Incomplete Student assignment linkages and DSP communications, held OH and taught sections
- Lead Teaching Assistant - EECS 122: Communication Networking/Internet:** Lead all administrative course logistics, **created** all existing course **content**, held office hours and **taught section** on-campus
- Reader, Academic Student Employee:** Graded (and wrote) exams on Data Structures (Lists/HashMaps/Heaps/Trees/etc) & Discrete Math (Logic, Crypto, Markov Chains, etc) & held Office Hours for **800+ students**
- Berkeley Research - Artificial Intelligence:** Deep Learning & **NLP:** Trained an **LSTM** to predict student grades from attendance and analyzed **Time Series** to determine earnestness with new SOTA **metrics** (Published a paper to be presented at ACM's 2023 conference). Fine-tuned a **LLM** on past TA messages on a Q&A Platform.

PROJECTS

- GPU-accelerated Factorization Machines:** Boosted recommender systems training speed by **21.4%** using a novel randomized algorithm that capitalized on data **sparsity** to parallelize Stochastic Gradient Descent. Achieved via custom CUDA structures for efficient **low-rank** polynomial kernel approximation. Profiled bottlenecks with VTune's Roofline.
- Berkeley AI Research (BAIR) Lab:** Collaborated with Simeon Adebola (PhD candidate) and Professor Goldberg from AUTOLab to develop a robust **automated** cable insertion system for **dynamic** environments. Applied CV and flood-fill algorithms for cable & channel identification using perception, segmentation, tracing & endpoint **detection**.

AWARDS

- Donald M. Rizzo Memorial Scholarship Recipient:** Awarded from international NPO: Lions Club International
- Winner of the Congressional App Challenge:** Invited to Capitol Hill to demo mobile app to House of Representative members at #HouseOfCode, the most prestigious student STEM expo, at the U.S. Capitol Building
- Capture The Flag Security Competitions:** Ranked **top 50** in all of {PicoCTF, HSCTF, BCACTF, CSAW Red}

SKILLS

- Languages:** Python, JavaScript, GoLang, Java, C, SQL, Typescript, Dart, R, C++11, CUDA, OCaml, x86, Lisp, C#
- Technologies:** Flask, Django, Flutter, React, React Native, Node.js, Firebase, PyTorch, Pandas, Git, RegEx, $\LaTeX$